Lecture 9

Example: 1

Minimize $f(x,y) = (x-2)^2+2(y-1)^2$
Subject to the constraint:

$$g_1(x,y) = x + 4y - 3 \leq 0$$
$$g_2(x,y) = y - x \leq 0$$

Define the **Lagrangian** as:

$L(x,y,\lambda_1,\lambda_2) = f(x,y) + \lambda_1 g_1(x,y) + \lambda_2 g_1(x,y)$

$L(x,y,\lambda_1,\lambda_2) = (x-2)^2+2(y-1)^2 + \lambda_1(x+4y-3) + \lambda_2(y-x)$

- Calcualte the gradient of **Lagrangian:**
- $\nabla L = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda_1} \\ \frac{\partial L}{\partial \lambda_2} \end{pmatrix} = \begin{pmatrix} 2(x-2) + \lambda_1 - \lambda_2 \\ 4(y-1) + 4\lambda_1 + \lambda_2 \\ g_1(x,y) \\ g_2(x,y) \end{pmatrix}$

- **Set up the KKT conditions:**
  - **1: Stationarity:**
  - $\nabla_x L(x^*, \lambda^*) = \mathbf{0}$
  - $\Rightarrow \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \end{pmatrix} = \begin{pmatrix} 2(x-2) + \lambda_1 - \lambda_2 \\ 4(y-1) + 4\lambda_1 + \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 2(x-2) + \lambda_1 - \lambda_2 = \mathbf{0} & (\mathbf{1}) \\ 4(y-1) + 4\lambda_1 + \lambda_2 = \mathbf{0} & (\mathbf{2}) \end{cases}$

2:Primal Feasibility::

- $g_1(x,y) = x + 4y - 3 \leq 0$
- $g_2(x,y) = y - x \leq 0$

3:Dual Feasibility:

$\lambda_1 \geq 0, \lambda_2 \geq 0$

Example:1 contd.

4: **Complementary Slackness:**

- $\lambda_1 g_1(x,y) = 0 \Longrightarrow \lambda_1(x + 4y - 3) = 0$
- $\lambda_2 g_2(x,y) = 0 \Longrightarrow \lambda_2(y - x) = 0$

Case Analysis: With respect to two constraints, four possible cases are:

- $\lambda_1 = 0, \lambda_2 = 0$: Both constraints are inactive i.e $A(x^*) = \emptyset$
- $\lambda_1 > 0, \lambda_2 = 0$: g_1 is active and g_2 is inactive i.e $A(x^*) = \{1\}$
- $\lambda_1 = 0, \lambda_2 > 0$: g_1 is inactive and g_2 is active i.e $A(x^*) = \{2\}$
- $\lambda_1 > 0, \lambda_2 > 0$: Both constraints are active i.e $A(x^*) = \{1,2\}$

Finally, With respect to the mentioned four cases, we have to find x, y, λ_1 and λ_2 which satisfy the KKT conditions:

- Minimize $f(x,y) = (x-2)^2 + 2(y-1)^2$
- $\begin{cases} 2(x-2) + \lambda_1 - \lambda_2 = 0 \\ 4(y-1) + 4\lambda_1 + \lambda_2 = 0 \end{cases}$ **Stationarity conditions**
- $\begin{cases} g_1(x,y) = x + 4y - 3 \leq 0 \\ g_2(x,y) = y - x \leq 0 \end{cases}$ **Primal Feasibility**
- $\begin{cases} \lambda_1 \geq 0 \\ \lambda_2 \geq 0 \end{cases}$ **Dual Feasibility**
- $\begin{cases} \lambda_1(x + 4y - 3) = 0 \\ \lambda_2(y - x) = 0 \end{cases}$ **Complementary Slackness**

Example:1 contd.

- **Case 1: ($\lambda_1 = 0, \lambda_2 = 0$) No active constraints**
  - $g_1(x,y) = x + 4y - 3 < 0$
  - $g_2(x,y) = y - x < 0$
- **Stationary conditions:**
  - From eqn.(1) and (2): $\begin{cases} 2(x-2) + \lambda_1 - \lambda_2 = 0 \\ 4(y-1) + 4\lambda_1 + \lambda_2 = 0 \end{cases} \Rightarrow \begin{cases} 2(x-2) = 0 \\ 4(y-1) = 0 \end{cases} \Rightarrow \begin{cases} x = 2 \\ y = 1 \end{cases}$
  - Now $g_1(x,y) = x + 4y - 3 \Rightarrow g_1(2,1) = 2 + 4.1 - 3 = 3 > 0$ **Contradiction** since primal feasibility: $g_1(x,y) \leq 0$

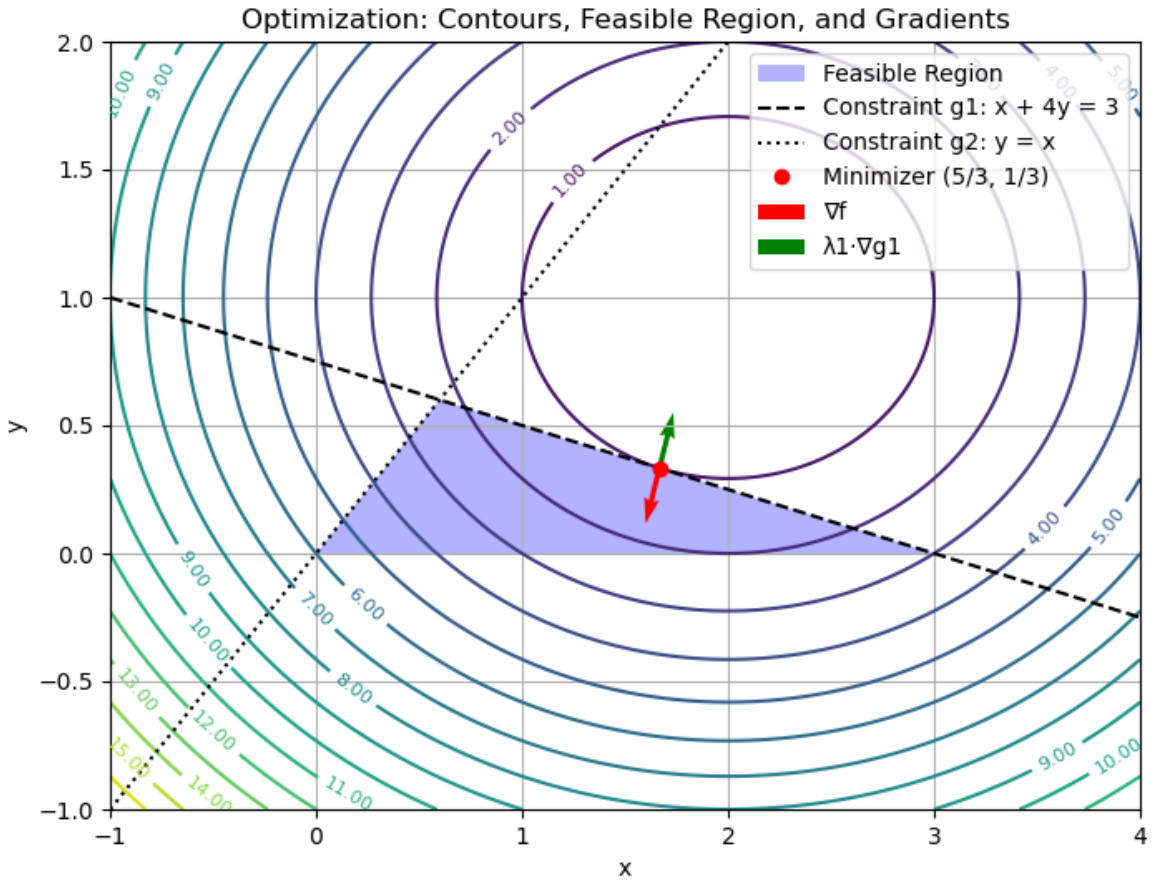
- **Case 2: Only g_1 is active** ($\lambda_1 > 0$ **and** $\lambda_2 = 0$)**:**
  - $g_1(x,y) = x + 4y - 3 = 0$
  - $g_2(x,y) = y - x < 0$
- **Stationary conditions:**
  - From eqn.(1) and (2): $\begin{cases} 2(x-2) + \lambda_1 - \lambda_2 = 0 \\ 4(y-1) + 4\lambda_1 + \lambda_2 = 0 \end{cases} \Rightarrow \begin{cases} 2(x-2) + \lambda_1 = 0 \\ 4(y-1) + 4\lambda_1 = 0 \end{cases}$
  - Now we have three equations: $\begin{cases} 2(x-2) + \lambda_1 = 0 \\ 4(y-1) + 4\lambda_1 = 0 \\ x + 4y - 3 = 0 \end{cases} \Rightarrow x = \frac{5}{3}, y = \frac{1}{3}, \lambda_1 = \frac{2}{3}$
  - Now $g_1(x,y) = x + 4y - 3 \Rightarrow g_1\left(\frac{5}{3}, \frac{1}{3}\right) = \frac{5}{3} + \frac{4.1}{3} - 3 = 0$ satisfied primal feasibility
  - $g_2(x,y) = y - x \Rightarrow g_1\left(\frac{5}{3}, \frac{1}{3}\right) = \frac{1}{3} - \frac{5}{3} = -\frac{4}{3} < 0$ satisfied primal feasibility
  - Since $\lambda_1 = \frac{2}{3} \geq 0, \lambda_2 = 0$ Both satisfied dual feasibility
- Therefore, Case 2 is the potential solution (**KKT point**).

Example:1 contd.

- **Case 3: Only g_2 is active ($\lambda_1 = 0$ and $\lambda_2 > 0$):**
  - $g_1(x, y) = x + 4y - 3 < 0$
  - $g_2(x, y) = y - x = 0$
- **Stationary conditions:**
  - From eqn.(1) and (2): $\begin{cases} 2(x-2) + \lambda_1 - \lambda_2 = 0 \\ 4(y-1) + 4\lambda_1 + \lambda_2 = 0 \end{cases} \Rightarrow \begin{cases} 2(x-2) - \lambda_2 = 0 \\ 4(y-1) + \lambda_2 = 0 \end{cases}$
  - Now we have three equations: $\begin{cases} 2(x-2) - \lambda_2 = 0 \\ 4(y-1) + \lambda_2 = 0 \\ y - x = 0 \end{cases} \Rightarrow x = \frac{4}{3}, y = \frac{4}{3}, \lambda_2 = -\frac{4}{3}$ (Violated dual feasibility)

- **Case 4:** Both constraints are active ($\lambda_1 > 0$ and $\lambda_2 > 0$)**:**
  - $g_1(x, y) = x + 4y - 3 = 0$
  - $g_2(x, y) = y - x = 0$
- **Stationary conditions:**
  - From eqn.(1) and (2): $\begin{cases} 2(x-2) + \lambda_1 - \lambda_2 = 0 \\ 4(y-1) + 4\lambda_1 + \lambda_2 = 0 \end{cases}$
  - Now we have four equations: $\begin{cases} 2(x-2) + \lambda_1 - \lambda_2 = 0 \\ 4(y-1) + 4\lambda_1 + \lambda_2 = 0 \\ x + 4y - 3 = 0 \\ y - x = 0 \end{cases} \Rightarrow x = \frac{3}{5}, y = \frac{3}{5}, \lambda_1 = 0.88, \lambda_2 = -1.92$ (Violated dual feasibility)

Example:1 contd.



Example 2:

Minimize $f(x,y) = x^2 + y^2 + 1$
s.t. $g_1(x,y) = -x - y + 1 \leq 0$
$g_2(x,y) = x - y - 2 = 0$

$\nabla g_1(x,y)$ and $\nabla g_2(x,y)$ are linearly independent: Constraint qualification is well defined

Define the **Lagrangian** as:
$L(x,y,\lambda,\mu) = f(x,y) + \lambda g_1(x,y) + \mu g_1(x,y)$
$L(x,y,\lambda,\lambda_2) = x^2 + y^2 + 1 + \lambda(-x - y + 1) + \mu(x - y - 2)$

- Calcualte the gradient of **Lagrangian:**
- $\nabla L = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda} \\ \frac{\partial L}{\partial \mu} \end{pmatrix} = \begin{pmatrix} 2x - \lambda + \mu \\ 2y - \lambda - \mu \\ g_1(x,y) \\ g_2(x,y) \end{pmatrix}$

- **Set up the KKT conditions:**
  - **1: Stationarity:**
  - $\nabla_{x,y} L(x,y,\lambda,\mu) = \mathbf{0}$
  - $\Rightarrow \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \end{pmatrix} = \begin{pmatrix} 2x - \lambda + \mu \\ 2y - \lambda - \mu \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} 2x - \lambda + \mu = \mathbf{0} & (\mathbf{1}) \\ 2y - \lambda - \mu = \mathbf{0} & (\mathbf{2}) \end{cases}$

2:Primal Feasibility::
- $g_1(x,y) = -x - y + 1 \leq 0$
- $g_2(x,y) = x - y - 2 = 0$

3:Dual Feasibility:
$\lambda_1 \geq 0, \mu \in \mathbb{R}$

Example 2: Contd.

4: **Complementary Slackness:**

- $\lambda g_1(x,y) = 0 \Longrightarrow \lambda(-x-y+1) = 0$

Case Analysis: With respect to one constraints, two possible cases are:

- $\lambda = 0$: g_1 is inactive i.e., $A(x^*) = \{g_2\}$
- $\lambda > 0$: g_1 is active i.e., $A(x^*) = \{g_1, g_2\}$
- Since $g_2(x,y)$ is an equality constraints therefore, in both above cases, $g_2(x,y)$ is active by default with $\mu \in \mathbb{R}$

Finally, With respect to the mentioned two cases, we have to find x, y, λ and μ which satisfy the KKT conditions:

- Minimize $f(x,y) = x^2 + y^2 + 1$
- $\begin{cases} 2x - \lambda + \mu = 0 \\ 2y - \lambda - \mu = 0 \end{cases}$ **Stationarity conditions**
- $\begin{cases} -x - y + 1 \leq 0 \\ x - y - 2 = 0 \end{cases}$ **Primal Feasibility**
- $\begin{cases} \lambda \geq 0 \\ \mu \in \mathbb{R} \end{cases}$ **Dual Feasibility**
- $\lambda(-x-y+1) = 0$ **Complementary Slackness**

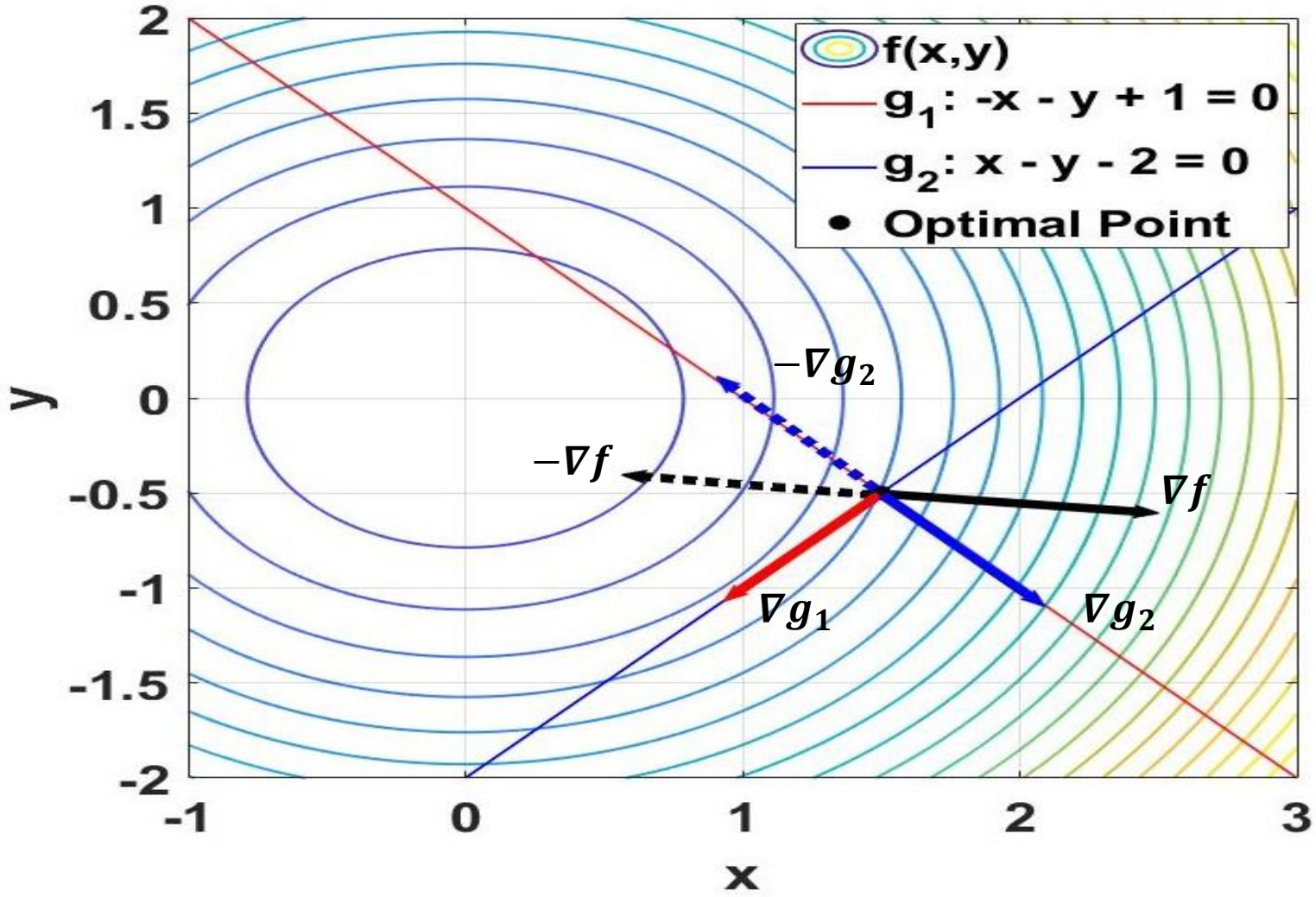
Example: Contd.

- **Case 1: ($\lambda = 0$) g_1 is inactive constraint, g_2 is active constraint (by default) with $\mu \in \mathbb{R}$**
  - $g_1(x, y) = -x - y + 1 < 0$
  - $g_2(x, y) = x - y - 2 = 0$
- **Stationary conditions:**
  - From eqn.(1) and (2): $\begin{cases} 2x - \lambda + \mu = 0 \\ 2y - \lambda - \mu = 0 \end{cases} \Rightarrow \begin{cases} 2x + \mu = 0 \\ 2y - \mu = 0 \end{cases} \Rightarrow \begin{cases} x = -\frac{\mu}{2} \\ y = \frac{\mu}{2} \end{cases}$
  - Now $g_1(x, y) = -x - y + 1 \Rightarrow g_1\left(-\frac{\mu}{2}, \frac{\mu}{2}\right) = \frac{\mu}{2} - \frac{\mu}{2} + 1 < 0 \Rightarrow \mathbf{1 < 0}$ **Contradiction**

- **Case 1: ($\lambda > 0$) g_1 is active constraint, g_2 is active constraint (by default) with $\mu \in \mathbb{R}$**
  - $g_1(x, y) = -x - y + 1 = 0$
  - $g_2(x, y) = x - y - 2 = 0$
- **Stationary conditions:**
  - From eqn.(1) and (2): $\begin{cases} 2x - \lambda + \mu = 0 \\ 2y - \lambda - \mu = 0 \end{cases}$
  - Now we have four equations: $\begin{cases} 2x - \lambda + \mu = 0 \\ 2y - \lambda - \mu = 0 \\ -x - y + 1 = 0 \\ x - y - 2 = 0 \end{cases} \Rightarrow x = \frac{3}{5}, y = -\frac{1}{2}, \lambda = 1, \mu = -2$

Now **KKT point** is $f\left(\frac{3}{5}, -\frac{1}{2}\right) = \left(\frac{3}{5}\right)^2 + \left(-\frac{1}{2}\right)^2 + 1 = \frac{7}{2}$

Example 2: Contd.



Example 3:

Minimize $\min_{x \in R^2} f(x) = \| x - x^* \|_2^2, x = [x, y]^T, x^* = [3, 2]^T$

s.t. $h(x, y) = x + y - 4 = 0$

$g(x, y) = x \leq 0$

Minimize $\min_{x \in R^2} f(x) = (x - 3)^2 + (y - 2)^2$

s.t. $h(x, y) = x + y - 4 = 0$

$g(x, y) = x \leq 0$

Second order optimality conditions:

- If you're at a point x^* satisfying the KKT conditions (x^* may be maximum and saddle point), to confirm that it's a local minimizer, you look at the second derivative (Hessian of the Lagrangian).
- But you can't check positive definiteness everywhere — only in directions you can legally move (the Critical Cone).
- **Second order sufficient condition:**
  - **If $d^T \nabla^2_{xx} L(x^*, \lambda^*, \mu^*) d > 0$ for all $d \in C(x^*) \backslash \{0\}$, then x^* is strict local minimum**
    - *The matrix $\nabla^2_{xx} L(x^*, \lambda^*, \mu^*)$ is the Hessian of the Lagrangian.*
    - *The condition does not require positive definiteness over all $\mathbb{R}^n$.*
    - *Instead, it requires positive definiteness restricted to the critical cone.*

- Tangent cone → where you can go
- Critical cone → where first-order gives no information
- Hessian → what happens next (curvature)

- For a convex constrained optimization problem, if x^*satisfies the KKT conditions, then x^* is a global minimizer. Therefore, **no second-order test is needed to establish global optimality**